

# MUTUAL FUND ANALYTICS PLATFORM USING PYTHON, SQLITE AND POWER BI

## Final Internship Capstone Project Report

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**Submitted To:** Bluestock Fintech

### Technologies Used:

Python | SQLite | Power BI | GitHub | VS Code

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### Executive Summary

The objective of this project was to design and implement a comprehensive Mutual Fund Analytics Platform capable of processing large-scale mutual fund datasets and transforming them into meaningful business insights for investors and financial analysts.

The solution integrates data engineering, financial analytics, and business intelligence techniques through a complete ETL (Extract, Transform, Load) pipeline. Multiple mutual fund datasets were collected, cleaned, transformed, and loaded into a centralized SQLite database. Advanced analytical techniques were then applied to evaluate mutual fund performance, investor behavior, sector allocation, and risk-adjusted returns.

A Power BI dashboard was developed to provide interactive visualizations and decision-support tools. Additional functionalities such as live NAV integration and a risk-based mutual fund recommendation engine were implemented to enhance practical usability.

The final platform demonstrates how data-driven analytics can improve investment decision-making and support investor engagement in the mutual fund ecosystem.

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### Chapter 1: Introduction

## **1.1 Background**

Mutual funds have become one of the most preferred investment instruments among retail investors due to diversification, professional management, and accessibility. However, with thousands of schemes available across categories, selecting suitable funds can be challenging.

Investors require analytical tools capable of evaluating performance, risk, and investment suitability based on objective metrics.

## **1.2 Problem Statement**

Investors often face difficulties in:

- Comparing multiple mutual funds.
- Evaluating risk-adjusted returns.
- Understanding fund category trends.
- Monitoring investment performance.
- Making informed investment decisions.

## **1.3 Project Objectives**

The objectives of this project are:

- Build a complete ETL pipeline.
- Design a SQLite analytical database.
- Analyze mutual fund performance.
- Develop investor insights.
- Create Power BI dashboards.
- Implement live NAV integration.
- Develop a recommendation engine.

## **1.4 Scope of Project**

The project focuses on:

- Historical mutual fund data analysis.
- Risk analytics.
- Performance benchmarking.
- Investor behavior analytics.

- Interactive reporting.
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## **2.1 Fund Master Dataset**

Contains:

- Scheme Code
- Scheme Name
- Fund House
- Category
- Sub Category
- Risk Grade

## **2.2 NAV History Dataset**

Contains historical Net Asset Values used for:

- Daily Returns
- Volatility
- Sharpe Ratio
- Performance Analysis

## **2.3 AUM Dataset**

Contains Assets Under Management data across multiple fund houses and categories.

## **2.4 Investor Transaction Dataset**

Contains:

- Purchase Transactions
- Redemption Transactions
- SIP Activity

## **2.5 Benchmark Dataset**

Contains benchmark indices used for fund comparison.

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## **Chapter 3: System Architecture**

### 3.1 Overall Architecture

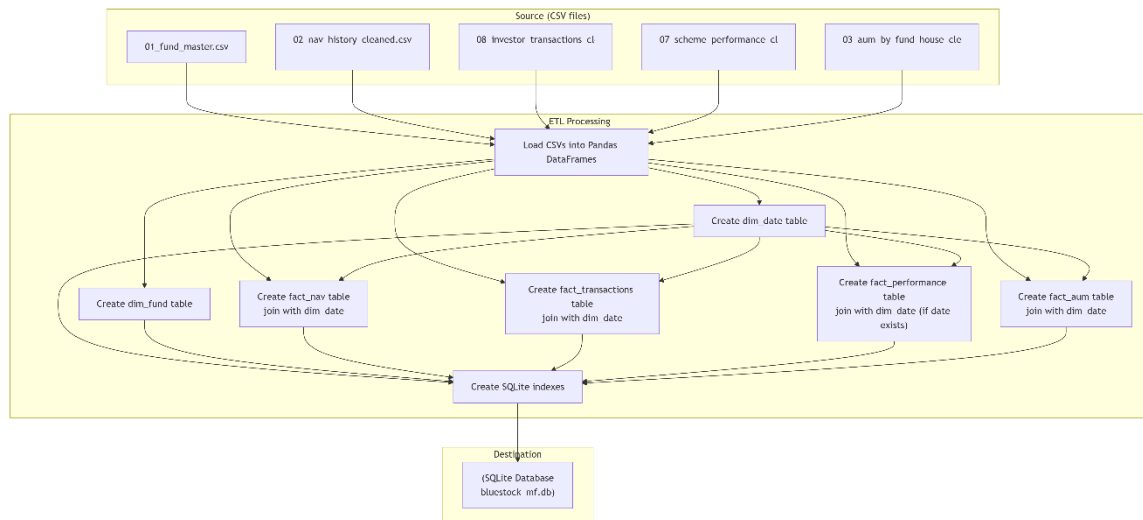


Figure 1: Overall ETL and Analytics Architecture

#### Workflow

Raw CSV Files



Data Cleaning



Feature Engineering



SQLite Database



Power BI Dashboard



Investor Insights

### 3.2 Technology Stack

| Component            | Technology                |
|----------------------|---------------------------|
| Programming Language | Python                    |
| Data Processing      | Pandas                    |
| Database             | SQLite                    |
| Dashboard            | Power BI                  |
| IDE                  | VS Code, Jupyter Notebook |
| Version Control      | Git & GitHub              |

|                       |         |
|-----------------------|---------|
| API Integration       | MFAPI   |
| Database Connectivity | sqlite3 |

### 3.3 Database Design

Dimension Tables:

- dim\_fund
- dim\_date

Fact Tables:

- fact\_nav
- fact\_transactions
- fact\_performance
- fact\_aum

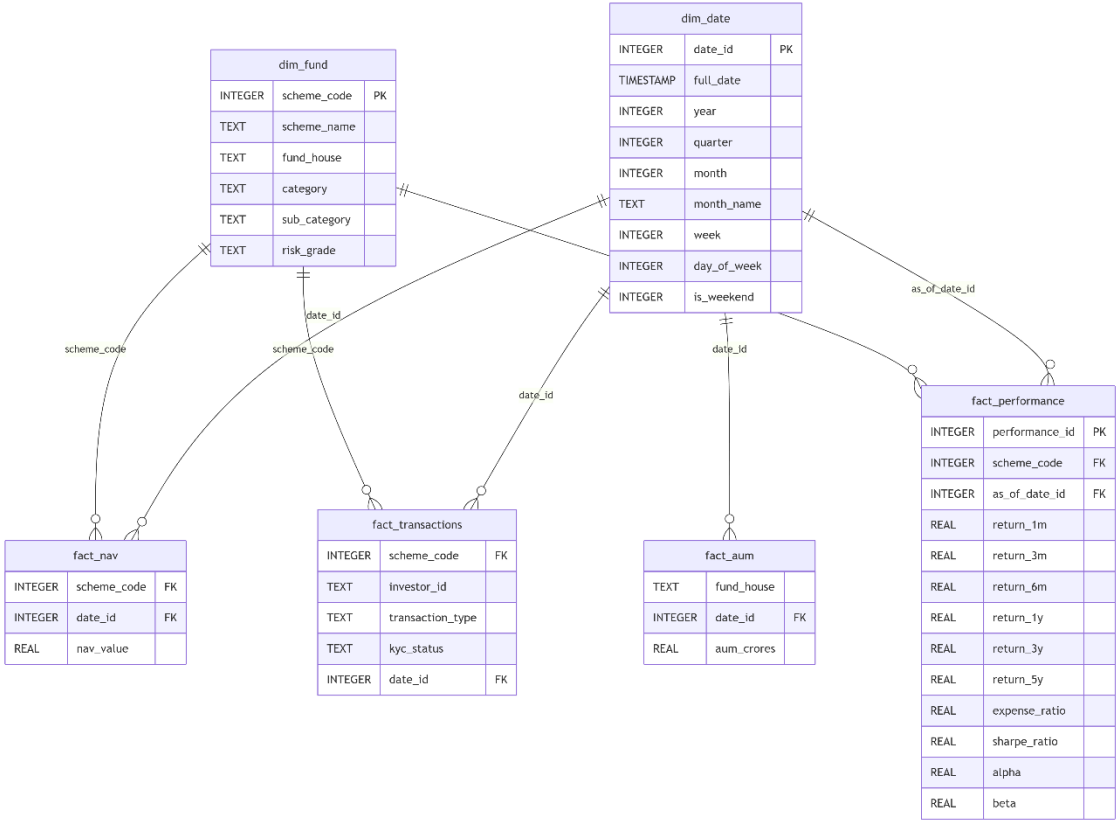


Figure 2: SQLite Database Schema

## Chapter 4: ETL Pipeline Design

### 4.1 Data Extraction

Data was extracted from multiple CSV files containing historical mutual fund information.

### 4.2 Data Cleaning

Cleaning operations included:

- Missing value handling
- Duplicate removal
- Date standardization
- Data type corrections

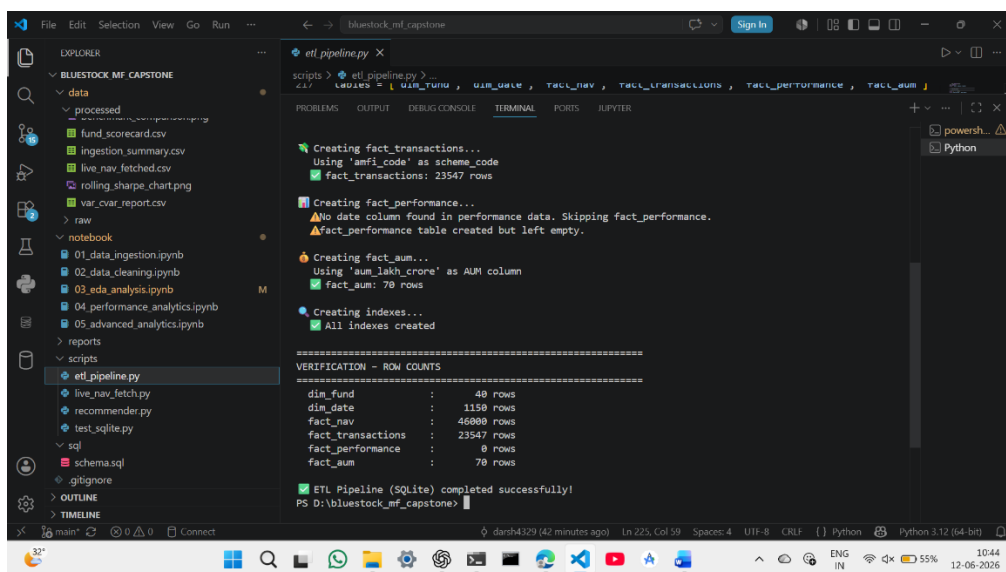
### 4.3 Data Transformation

Transformation steps included:

- Daily return calculation
- Category aggregation
- Benchmark normalization
- Risk metric computation

### 4.4 Data Loading

Processed data was loaded into SQLite using Python's sqlite3 module.



```
scripts > etl_pipeline.py > ...  
### tables = [ dim_fund , dim_date , fact_nav , fact_transactions , fact_performance , fact_aum ]  
  
Creating fact_transactions...  
Using 'amfi_code' as schema_code  
fact_transactions: 23547 rows  
  
Creating fact_performance...  
No date column found in performance data. Skipping fact_performance.  
fact_performance table created but left empty.  
  
Creating fact_aum...  
Using 'aum_lakh_crore' as AUM column  
fact_aum: 70 rows  
  
Creating indexes...  
All indexes created  
  
===== VERIFICATION - ROW COUNTS =====  
dim_fund      : 40 rows  
dim_date     : 1150 rows  
fact_nav     : 46000 rows  
fact_transactions : 23547 rows  
fact_performance : 0 rows  
fact_aum     : 70 rows  
  
ETL Pipeline (SQLite) completed successfully!  
PS D:\bluestock_mf_capstone>
```

Figure 3: ETL Pipeline Execution

Chapter 5: Exploratory Data Analysis

5.1 AUM Growth Analysis

[INSERT AUM GROWTH BY FUND HOUSE CHART HERE]

Caption:  
Figure 4: AUM Growth by Fund House (2022–2025)

Key Findings

- SBI Mutual Fund recorded highest AUM.
- ICICI Prudential demonstrated strong growth.
- HDFC maintained consistent expansion.

5.2 Category Inflow Analysis

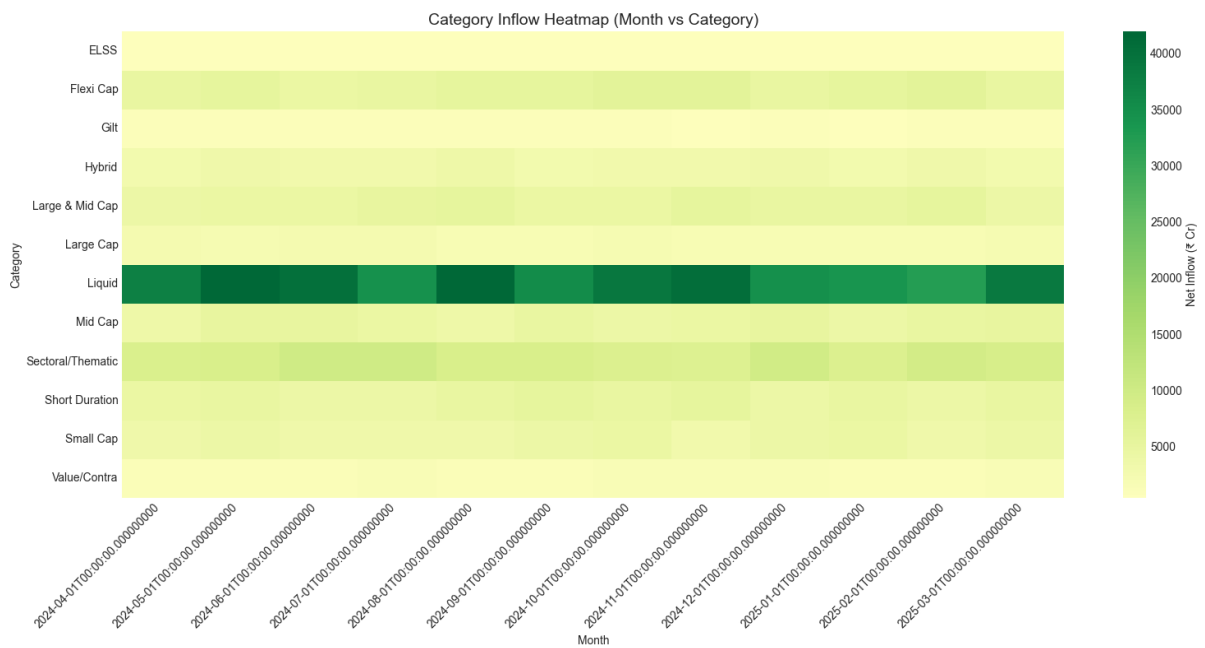


Figure 5: Category-wise Monthly Inflows

Key Findings

- Liquid Funds attracted maximum inflows.
- Flexi Cap and Sectoral Funds showed steady growth.
- ELSS category remained relatively stable.

5.3 Return Distribution Analysis



### Daily Return Distribution (5 sample funds)

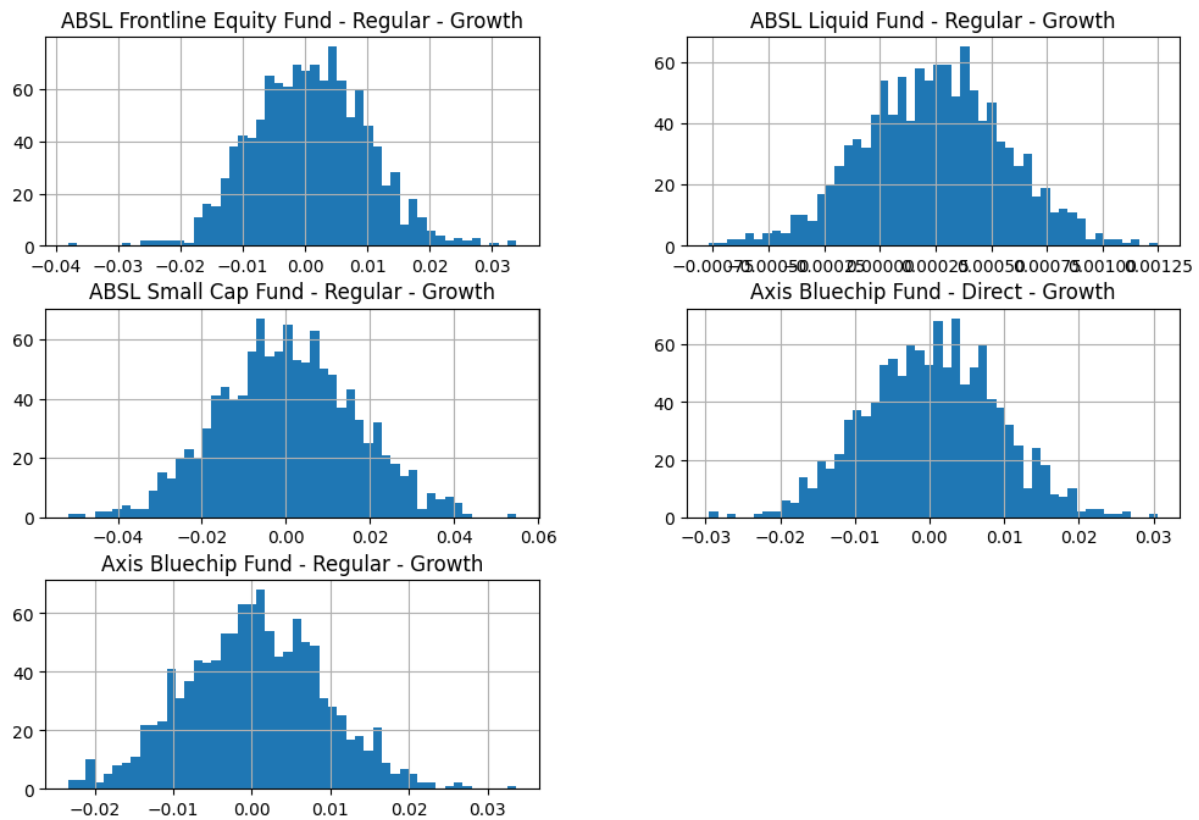


Figure 6: Distribution of Daily Returns

#### Key Findings

- Equity funds exhibit higher volatility.
- Liquid funds demonstrate stability.
- Small-cap funds show wider return ranges.

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## 5.4 Correlation Analysis

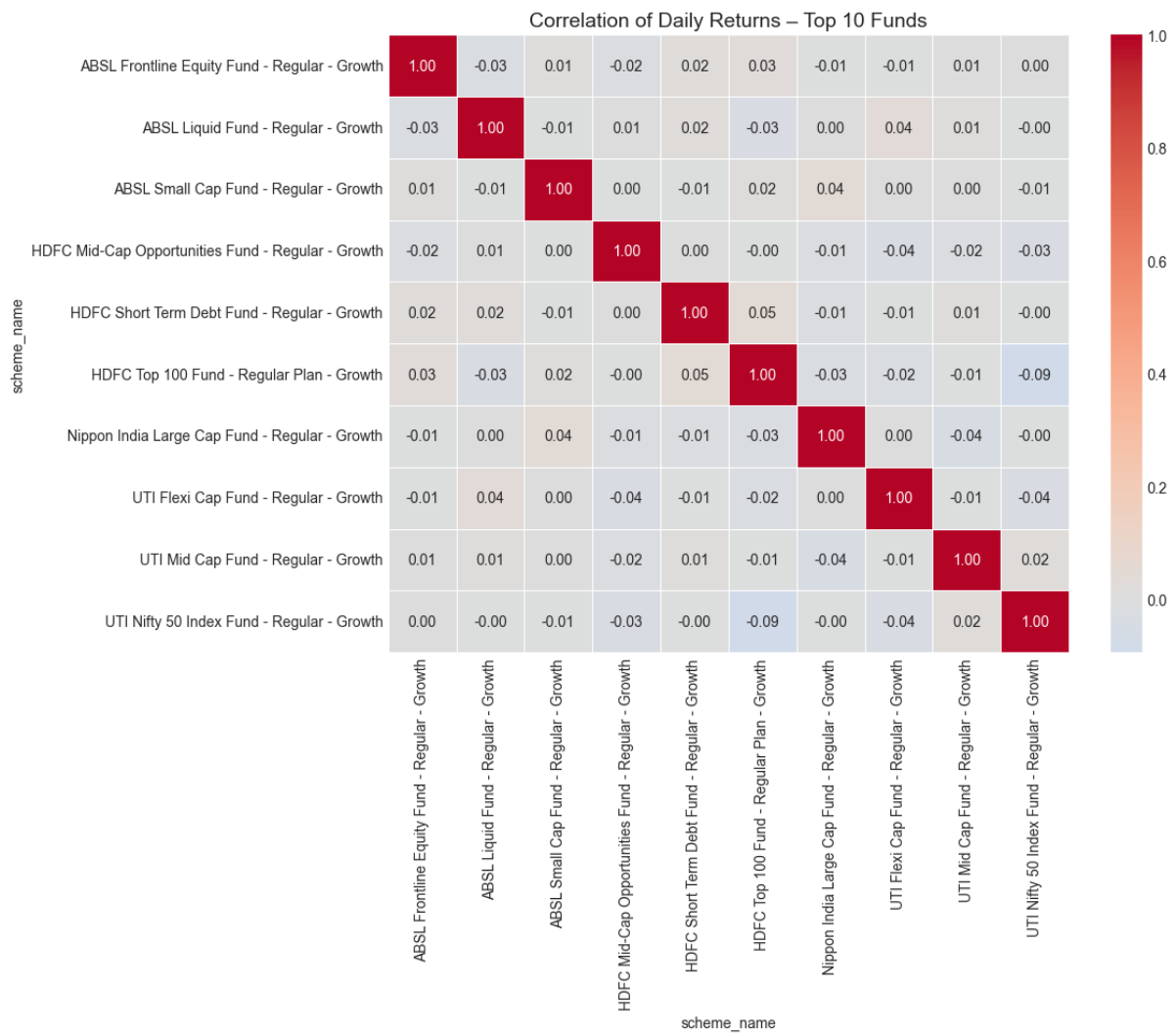


Figure 7: Correlation Matrix of Top 10 Funds

### Key Findings

- Most funds exhibit low-to-moderate correlations.
- Diversification opportunities exist across categories.

## Chapter 6: Performance Analytics

### 6.1 Sharpe Ratio Analysis

The Sharpe Ratio was used to evaluate risk-adjusted returns.

$$\text{Sharpe Ratio} = \frac{\text{Return} - \text{Risk Free Rate}}{\text{Standard Deviation}}$$

#### Interpretation

Higher Sharpe Ratio indicates superior risk-adjusted performance.

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### 6.2 Rolling Sharpe Ratio

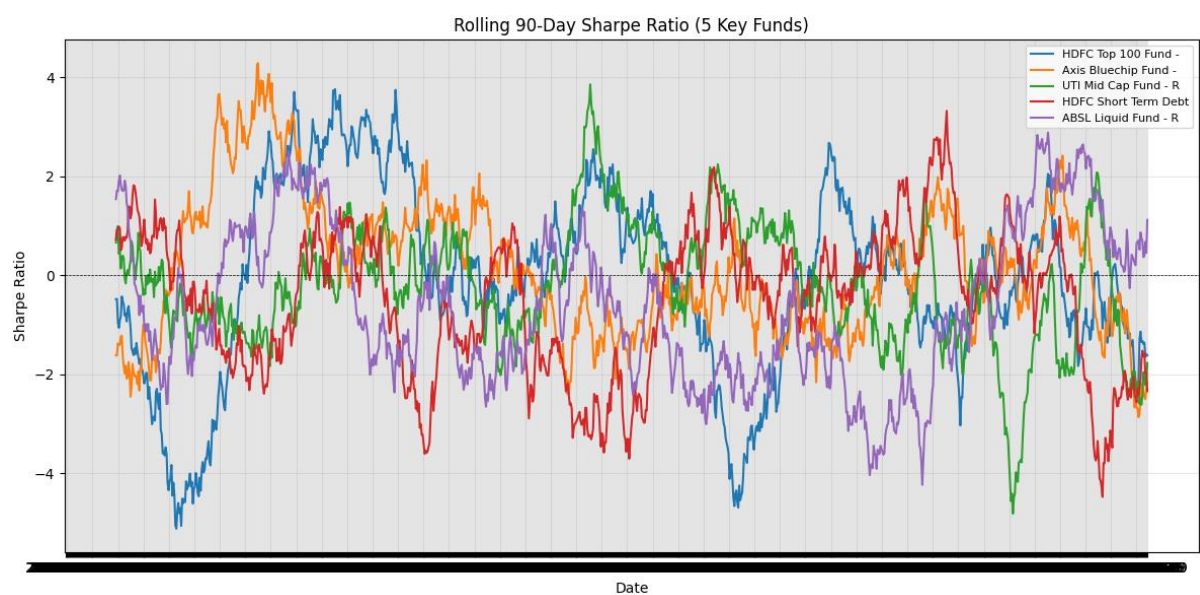


Figure 8: Rolling 90-Day Sharpe Ratio

#### Observations

- Positive Sharpe periods indicate strong performance.
- Volatility impacts risk-adjusted returns.

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### 6.3 Benchmark Comparison

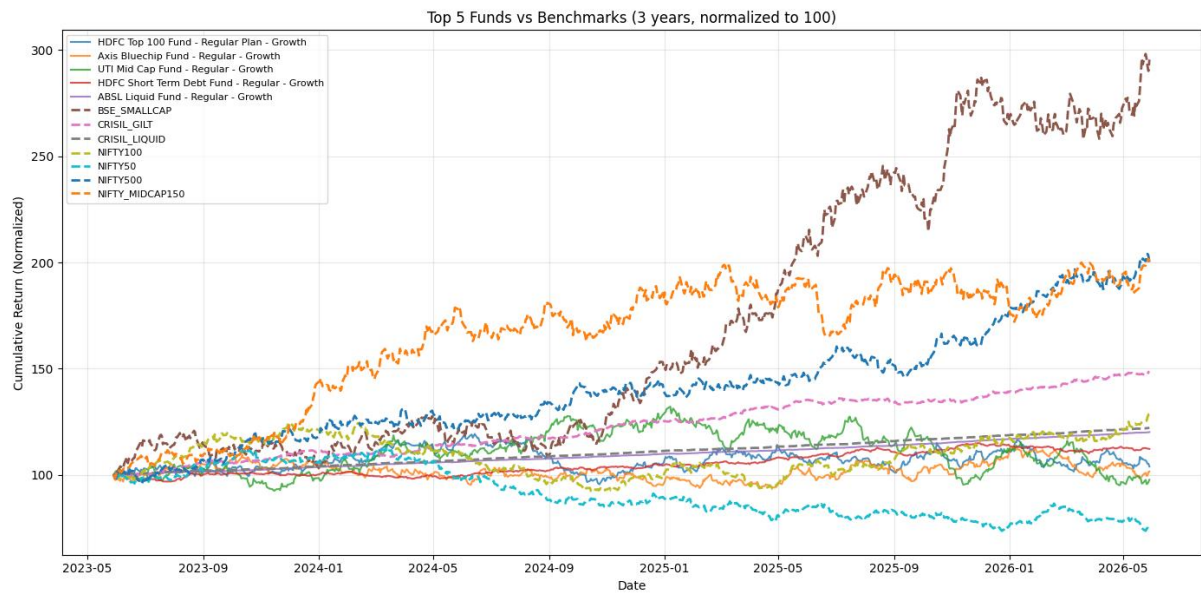


Figure 9: Top Funds vs Benchmark Indices

### Findings

- Several actively managed funds outperformed benchmarks.
- Benchmark comparison validates fund performance consistency.

## Chapter 7: Investor Analytics

### 7.1 Investor Transaction Analysis

Industry Folio Count Growth (Jan 2022 – Dec 2025)

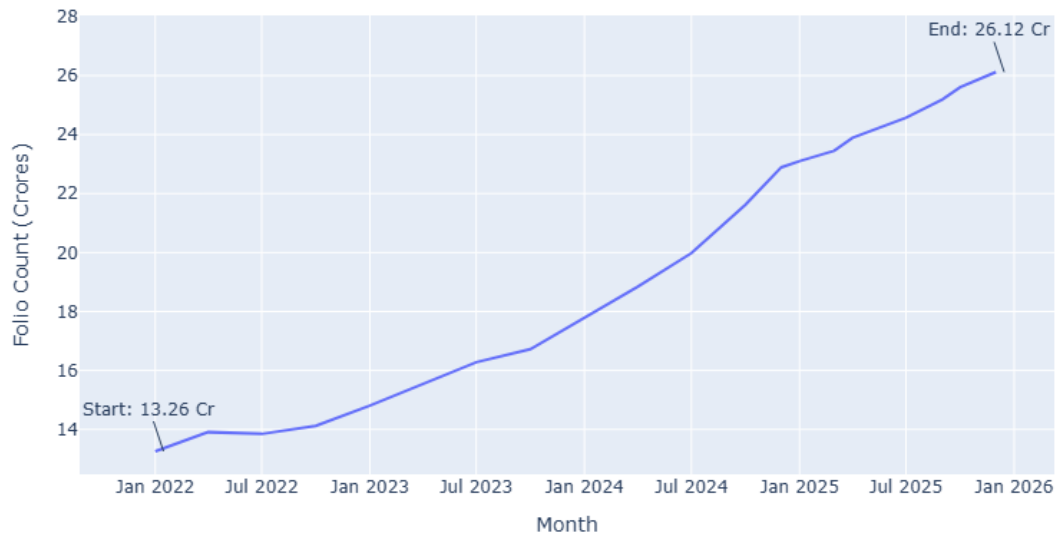


Figure 10: Investor Activity Trends

### 7.2 SIP Continuity Analysis

Monthly SIP Inflows (Jan 2022 – Dec 2025)

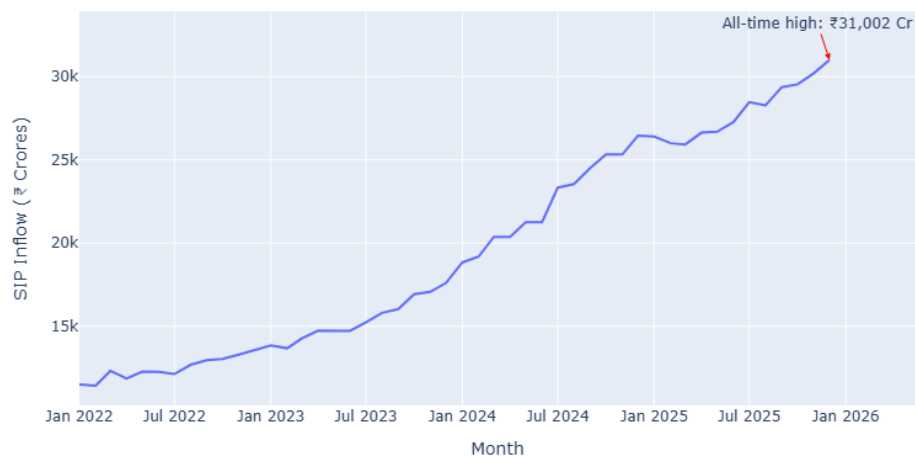


Figure 11: SIP Continuity Analysis

#### Findings

- Regular SIP investors show long-term investment discipline.
- SIP interruptions highlight investor churn opportunities.

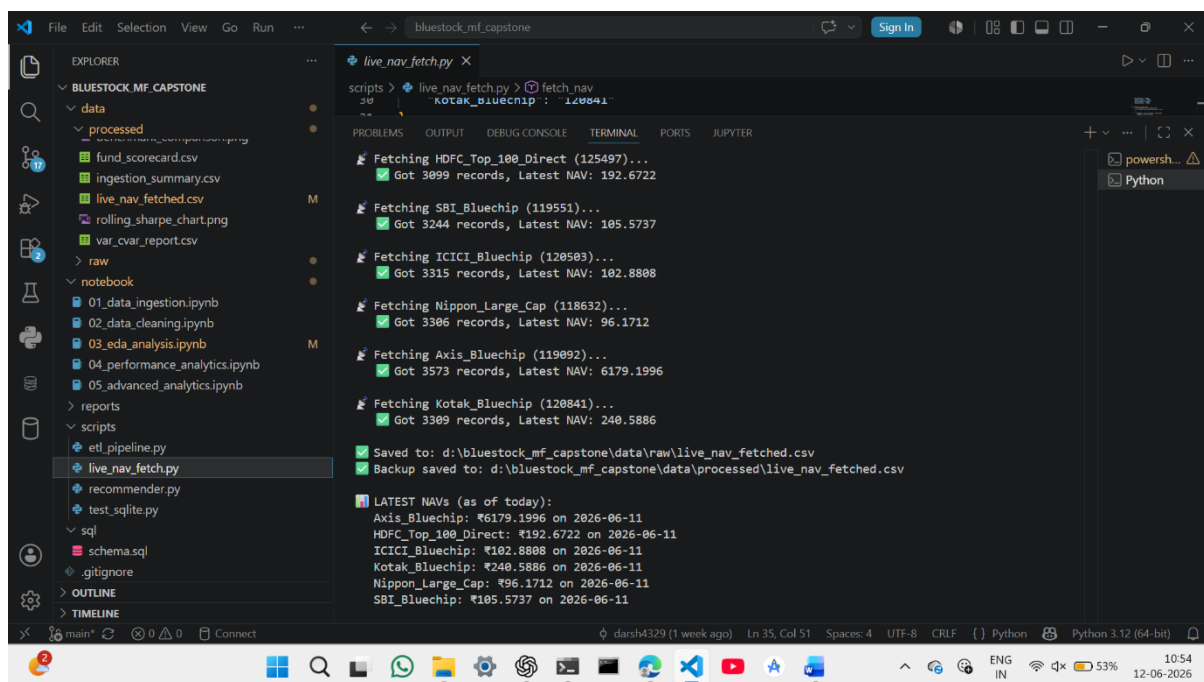
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## Chapter 8: Live NAV Integration

A dedicated Python script was developed to fetch live NAV data using MFAPI.

Features:

- Automated data retrieval.
- Historical NAV tracking.
- Data validation.
- CSV export.



```
live_nav_fetch.py X
scripts > live_nav_fetch.py > fetch_nav
kotak_bluechip: "120841"

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

Fetching HDFC_Top_100_Direct (125497)...
Got 3099 records, Latest NAV: 192.6722

Fetching SBI_Bluechip (119551)...
Got 3244 records, Latest NAV: 105.5737

Fetching ICICI_Bluechip (120503)...
Got 3315 records, Latest NAV: 102.8808

Fetching Nippon_Large_Cap (118632)...
Got 3306 records, Latest NAV: 96.1712

Fetching Axis_Bluechip (119092)...
Got 3573 records, Latest NAV: 6179.1996

Fetching Kotak_Bluechip (120841)...
Got 3309 records, Latest NAV: 240.5886

Saved to: d:\bluestock_mf_capstone\data\raw\live_nav_fetched.csv
Backup saved to: d:\bluestock_mf_capstone\data\processed\live_nav_fetched.csv

LATEST NAVs (as of today):
Axis_Bluechip: ₹6179.1996 on 2026-06-11
HDFC_Top_100_Direct: ₹192.6722 on 2026-06-11
ICICI_Bluechip: ₹102.8808 on 2026-06-11
Kotak_Bluechip: ₹240.5886 on 2026-06-11
Nippon_Large_Cap: ₹96.1712 on 2026-06-11
SBI_Bluechip: ₹105.5737 on 2026-06-11
```

Figure 12: Live NAV Data Fetching

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## Chapter 9: Recommendation Engine

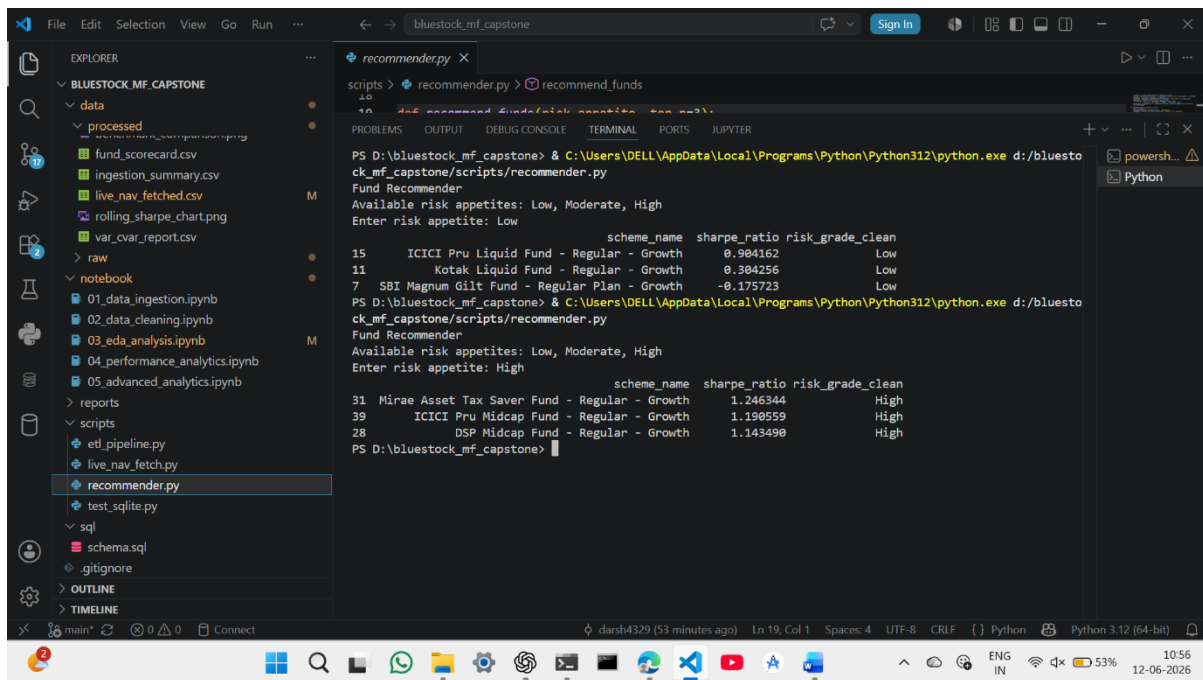
A risk-based recommendation engine was developed to assist investors.

Risk Profiles:

- Conservative
- Moderate
- Aggressive

Methodology:

- Risk Classification
- Sharpe Ratio Ranking
- Top Fund Selection



```
PS D:\bluestock_mf_capstone> & C:\Users\DELL\AppData\Local\Programs\Python\Python312\python.exe d:/bluestock_mf_capstone/scripts/recommender.py
Fund Recommender
Available risk appetites: Low, Moderate, High
Enter risk appetite: Low

  scheme_name  sharpe_ratio risk_grade_clean
15  ICICI Pru Liquid Fund - Regular - Growth    0.984162      Low
11  Kotak Liquid Fund - Regular - Growth       0.384256      Low
7   SBI Magnum Gilt Fund - Regular Plan - Growth -0.175723     Low
PS D:\bluestock_mf_capstone> & C:\Users\DELL\AppData\Local\Programs\Python\Python312\python.exe d:/bluestock_mf_capstone/scripts/recommender.py
Fund Recommender
Available risk appetites: Low, Moderate, High
Enter risk appetite: High

  scheme_name  sharpe_ratio risk_grade_clean
31  Mirae Asset Tax Saver Fund - Regular - Growth 1.246344      High
39  ICICI Pru Midcap Fund - Regular - Growth     1.190559      High
28  DSP Midcap Fund - Regular - Growth           1.143490      High
PS D:\bluestock_mf_capstone>
```

Figure 13: Fund Recommendation Engine

## Chapter 10: Power BI Dashboard

### Dashboard Page 1 – Overview

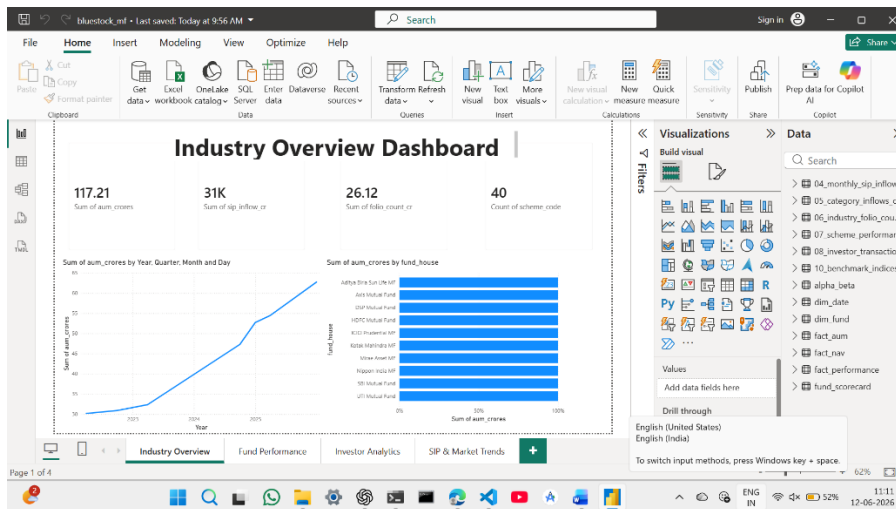


Figure 14: Overview Dashboard

### Dashboard Page 2 – NAV TREND

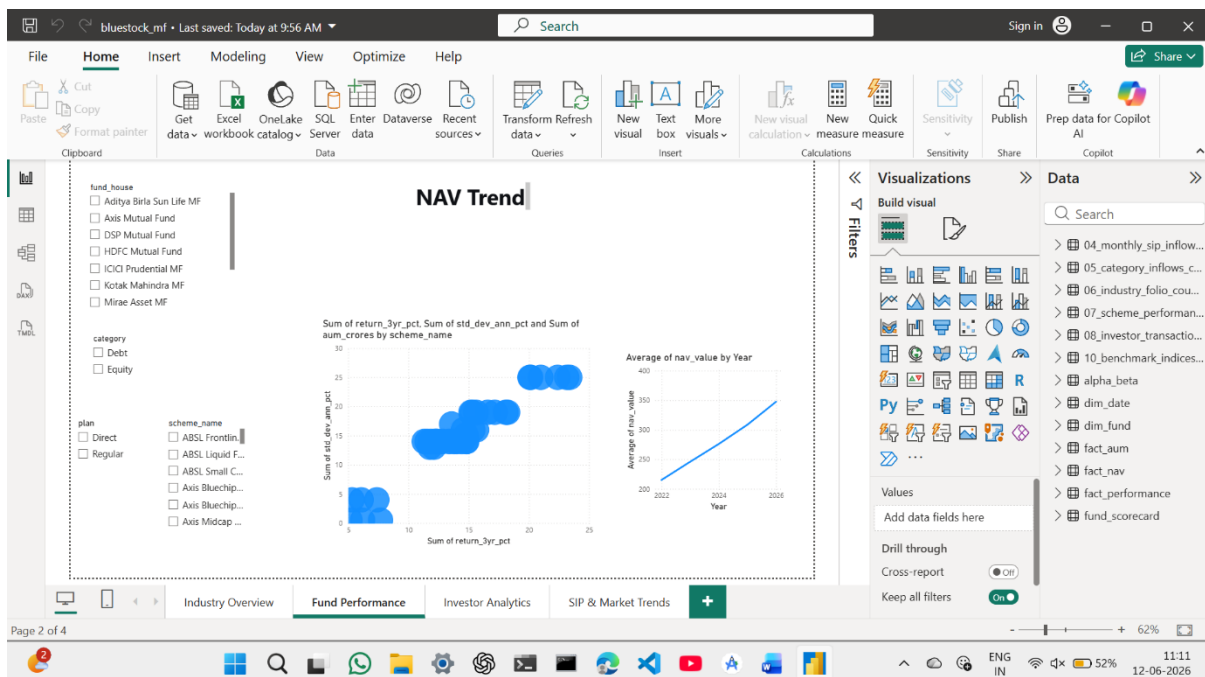


Figure 15:-NAV Dashboard

### Dashboard Page 3 – Investor Analytics



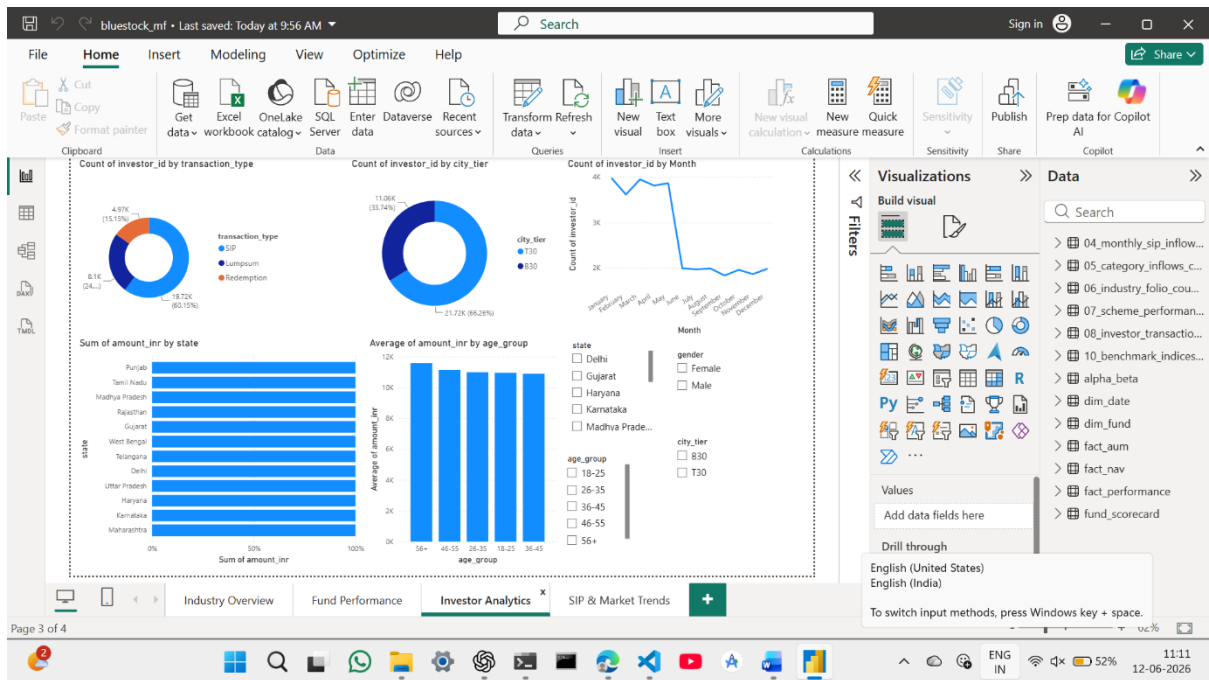


Figure 16: Investor Analytics Dashboard

## Dashboard Page 4 – SIP & Market Trends

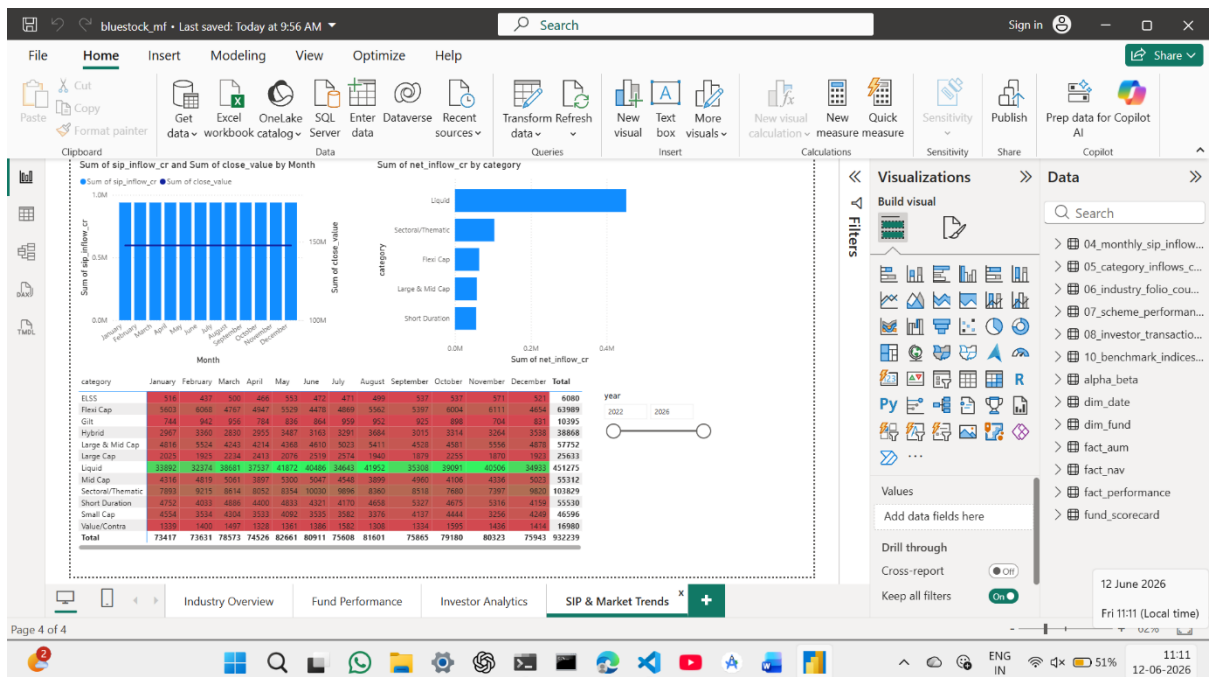


Figure 17: SIP & Market Trends Dashboard

## **Chapter 11: Results and Insights**

Key outcomes achieved:

- SQLite database successfully implemented.
  - More than 46,000 NAV records processed.
  - 40 mutual funds analyzed.
  - Risk-adjusted performance metrics generated.
  - Interactive Power BI dashboards deployed.
  - Recommendation engine developed.
- 

## **Chapter 12: Limitations**

The project has the following limitations:

- Historical analysis only.
  - Risk-free rate assumed as 0%.
  - Limited fund universe.
  - Mock transaction data.
  - No predictive forecasting.
- 

## **Chapter 13: Recommendations**

### **For Investors**

- Prioritize risk-adjusted returns.
- Diversify investments.
- Maintain SIP discipline.
- Monitor portfolio risk regularly.

### **For Bluestock**

- Introduce SIP interruption alerts.
- Implement predictive analytics.
- Expand recommendation engine.
- Add AI-powered investor assistance.

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## **Chapter 14: Conclusion**

This project successfully developed a complete Mutual Fund Analytics Platform using Python, SQLite, and Power BI. The system integrates ETL processing, financial analytics, investor behavior analysis, live NAV monitoring, and business intelligence reporting into a unified analytical framework.

The platform demonstrates the practical application of data engineering and analytics within the fintech sector and provides a scalable foundation for future enhancements.

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## **References**

1. SQLite Documentation
  2. Python Pandas Documentation
  3. Power BI Documentation
  4. MFAPI Documentation
  5. SEBI Mutual Fund Guidelines
  6. AMFI India Resources
- 

## **Appendix**

### **Appendix A – Project Folder Structure**

```
bluestock_mf_capstone/
├── data/
│   ├── raw/      ← original downloaded files
│   ├── processed/ ← cleaned, merged CSVs
│   └── db/        ← bluestock_mf.db (SQLite)
├── notebooks/
│   ├── 01_data_ingestion.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_eda_analysis.ipynb
│   ├── 04_performance_analytics.ipynb
│   └── 05_advanced_analytics.ipynb
├── scripts/
│   ├── etl_pipeline.py
│   ├── live_nav_fetch.py
│   ├── compute_metrics.py
│   └── recommender.py
├── sql/
│   ├── schema.sql
│   └── queries.sql
├── dashboard/
│   └── bluestock_mf.pbix
├── reports/
│   ├── Final_Report.pdf
│   └── Presentation.pptx
└── README.md
```

## Appendix B – Database Validation

| Table Name | Records |
|------------|---------|
|------------|---------|

|                   |        |
|-------------------|--------|
| dim_fund          | 40     |
| dim_date          | 1150   |
| fact_nav          | 46,000 |
| fact_transactions | 23547  |
| fact_aum          | 70     |

## Appendix C – GitHub Repository

The screenshot shows a web browser displaying the GitHub repository page for 'bluestock\_mf\_capstone' by user 'darsh4329'. The repository is on the 'main' branch and has 1 branch and 0 tags. The commit history shows a recent commit 'Update comment to include Sqlite in schema' by 'darsh4329' 1 hour ago. The file list includes 'dashboard', 'data', 'notebook', 'scripts', 'sql', '.gitignore', and 'requirements.txt'. The right sidebar shows the repository's 'About' section, 'Releases', 'Packages', and 'Contributors'.

**Repository Details:**

- Repository: bluestock\_mf\_capstone
- User: darsh4329
- Branch: main
- Commits: 7
- Files: dashboard, data, notebook, scripts, sql, .gitignore, requirements.txt
- Recent Commit: Update comment to include Sqlite in schema (1 hour ago)

**Right Sidebar:**

- About:** Mutual Fund Capstone Project
- Activity:** 0 stars, 0 watching, 0 forks
- Releases:** No releases published. [Create a new release](#)
- Packages:** No packages published. [Publish your first package](#)
- Contributors:** 1 contributor: darsh4329 Darsh